

```
import pandas as pd
```

```
sales=pd.read_csv('sales.csv')
```

```
sales.head(20)
```

	order_id	order_date	product_id	store_id	customer_id	quantity	unit_price	discount	revenue	cost	profit
0	ORD00000001	2023-01-07	P0080	S093	C040749	5	14.43	0.15	61.33	42.77	18.56
1	ORD00000002	2023-10-22	P0173	S065	C020161	3	12.01	0.00	36.03	19.06	16.97
2	ORD00000003	2023-05-07	P0115	S078	C048069	2	10.02	0.00	20.04	10.29	9.75
3	ORD00000004	2024-06-23	P0186	S088	C047901	2	14.66	0.10	26.39	16.35	10.04
4	ORD00000005	2024-09-24	P0197	S054	C033950	1	12.34	0.00	12.34	7.94	4.40
5	ORD00000006	2024-03-29	P0160	S089	C008918	4	13.52	0.00	54.08	36.59	17.49
6	ORD00000007	2023-02-26	P0062	S024	C002897	1	11.97	0.10	10.77	7.16	3.61
7	ORD00000008	2023-11-03	P0111	S085	C038072	5	4.62	0.00	23.10	16.15	6.95
8	ORD00000009	2024-10-11	P0135	S029	C003786	4	7.88	0.00	31.52	19.90	11.62
9	ORD00000010	2023-12-17	P0069	S056	C043148	3	8.88	0.00	26.64	18.19	8.45
10	ORD00000011	2023-01-01	P0181	S070	C016710	5	12.62	0.00	63.10	43.74	19.36
11	ORD00000012	2023-03-18	P0137	S028	C040620	4	4.10	0.00	16.40	9.90	6.50
12	ORD00000013	2024-11-06	P0003	S021	C048449	3	10.90	0.00	32.70	17.29	15.41
13	ORD00000014	2024-11-08	P0026	S005	C009995	2	4.43	0.10	7.97	5.50	2.48
14	ORD00000015	2024-01-08	P0042	S100	C011763	2	5.48	0.00	10.96	6.83	4.13
15	ORD00000016	2023-04-03	P0110	S037	C037715	2	5.93	0.00	11.86	6.11	5.75
16	ORD00000017	2024-08-16	P0058	S062	C038708	3	5.51	0.00	16.53	10.70	5.83
17	ORD00000018	2024-06-01	P0088	S069	C013225	5	8.54	0.00	42.70	26.15	16.55
18	ORD00000019	2024-03-26	P0069	S063	C001239	3	9.05	0.00	27.15	15.61	11.54
19	ORD00000020	2023-09-10	P0037	S049	C049903	1	4.52	0.10	4.07	2.22	1.84

```
from google.colab import drive
drive.mount('/content/drive')
```

```
products=pd.read_csv('products.csv')
```

```
stores=pd.read_csv('stores.csv')
```

```
products.head()
```

	product_id	product_name	brand	category	cocoa_percent	weight_g
0	P0001	White Chocolate 80%	Mars	Truffle	80	120
1	P0002	Dark Chocolate 70%	Cadbury	Praline	70	100
2	P0003	Truffle Chocolate 70%	Hershey	Praline	70	120
3	P0004	Milk Chocolate 50%	Mars	Praline	50	80
4	P0005	White Chocolate 70%	Ferrero	White	70	50

```
stores.head()
```

	store_id	store_name	city	country	store_type
0	S001	Chocolate Store 1	New York	Canada	Retail
1	S002	Chocolate Store 2	Melbourne	Canada	Mall
2	S003	Chocolate Store 3	Berlin	France	Mall
3	S004	Chocolate Store 4	Paris	UK	Airport
4	S005	Chocolate Store 5	Sydney	USA	Online

```
sales.head()
```

	order_id	order_date	product_id	store_id	customer_id	quantity	unit_price	discount	revenue	cost	profit
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4	ORD00000005	2024-09-24	P0197	S054	C033950	1	12.34	0.00	12.34	7.94	4.40

```
df=sales.merge(products,on='product_id')
```

```
df=df.merge(stores,on='store_id')
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 990236 entries, 0 to 990235
Data columns (total 20 columns):
#   Column              Non-Null Count  Dtype
---  -
0   order_id            990236 non-null object
1   order_date          990236 non-null object
2   product_id          990236 non-null object
3   store_id            990236 non-null object
4   customer_id         990236 non-null object
5   quantity            990236 non-null int64
6   unit_price          990236 non-null float64
7   discount            990236 non-null float64
8   revenue             990236 non-null float64
9   cost                990236 non-null float64
10  profit              990236 non-null float64
11  product_name        990236 non-null object
12  brand               990236 non-null object
13  category            990236 non-null object
14  cocoa_percent       990236 non-null int64
15  weight_g            990236 non-null int64
16  store_name          990236 non-null object
17  city                990236 non-null object
18  country             990236 non-null object
19  store_type          990236 non-null object
dtypes: float64(5), int64(3), object(12)
memory usage: 151.1+ MB
```

```
numericalcolumn=['quantity','unit_price','discount','revenue','cost','profit','cocoa_percent','weight_g']
```

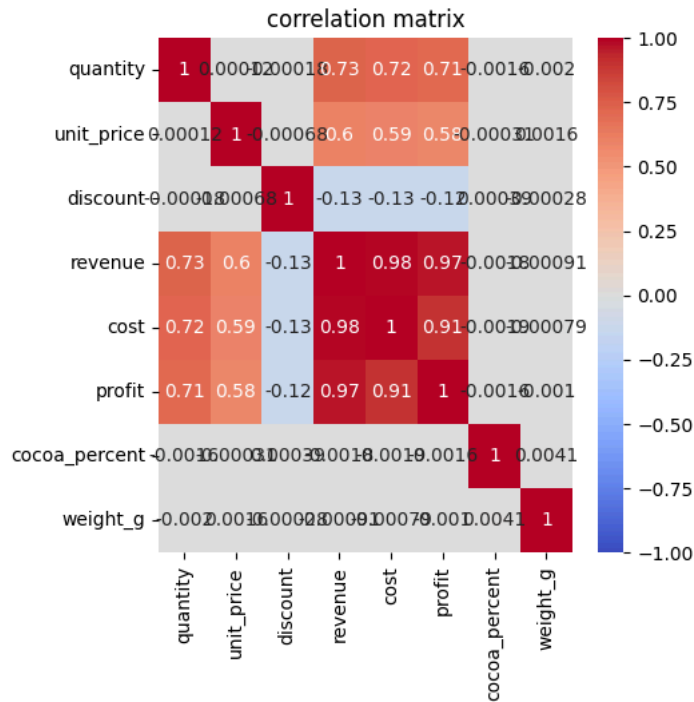
```
corrmat=df[numericalcolumn].corr()
```

```
print(corrmat)
```

	quantity	unit_price	discount	revenue	cost	profit	\
quantity	1.000000	0.000116	-0.000181	0.733963	0.722682	0.709035	
unit_price	0.000116	1.000000	-0.000681	0.599265	0.590006	0.578984	
discount	-0.000181	-0.000681	1.000000	-0.127325	-0.125151	-0.123319	
revenue	0.733963	0.599265	-0.127325	1.000000	0.984524	0.966194	
cost	0.722682	0.590006	-0.125151	0.984524	1.000000	0.906058	
profit	0.709035	0.578984	-0.123319	0.966194	0.906058	1.000000	
cocoa_percent	-0.001646	-0.000313	0.000391	-0.001842	-0.001926	-0.001616	
weight_g	-0.001952	0.001648	-0.000278	-0.000914	-0.000793	-0.001041	
	cocoa_percent	weight_g					
quantity	-0.001646	-0.001952					
unit_price	-0.000313	0.001648					
discount	0.000391	-0.000278					
revenue	-0.001842	-0.000914					
cost	-0.001926	-0.000793					
profit	-0.001616	-0.001041					
cocoa_percent	1.000000	0.004114					
weight_g	0.004114	1.000000					

```
import matplotlib.pyplot as plt
import seaborn as sns
```

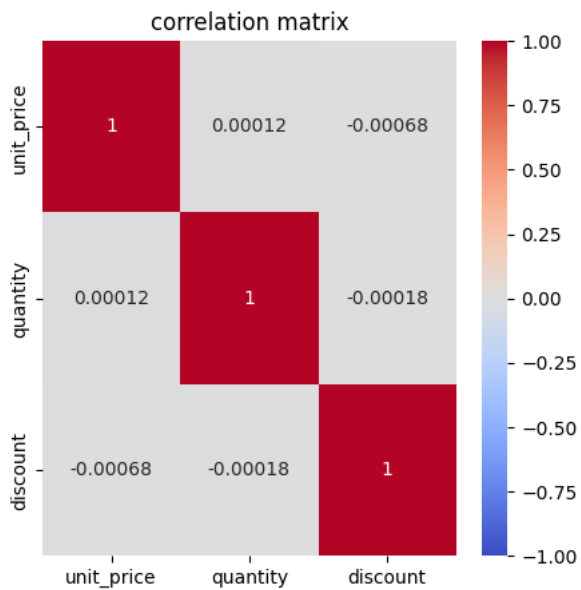
```
plt.figure(figsize=(5,5))
sns.heatmap(corrmat,annot=True,cmap='coolwarm',vmin=-1,vmax=1)
plt.title('correlation matrix')
plt.show()
```



```
numericalcolumn=['unit_price','quantity','discount']
```

```
corrmat=df[numericalcolumn].corr()
```

```
plt.figure(figsize=(5,5))
sns.heatmap(corrmat,annot=True,cmap='coolwarm',vmin=-1,vmax=1)
plt.title('correlation matrix')
plt.show()
```



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